



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and
Information Technology

EAI 320

INTELLIGENT SYSTEMS

PRACTICAL 4

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APPENDIX A CODE

```
1 % Intelligent Systems EAI 320 Practical 4 (NAive Bayes)
2 % Moosa Osman u24566722
3 % Last updated: 20260516
4
5 function rps = agent_bayes(previous, n_rounds)
6     persistent opponentHistory;
7     persistent ourHistory;
8
9     % General procedure:
10    % Random for move one ==> start predicting based on previous
11    ↪ moves
12    %More moves ==> greater chance of predicting mmove properly
13    % perform Naive Bayes and find most likely mvoe, then choose
14    ↪ winning move
15
16    if isempty(previous)
17        opponentHistory = [];
18        ourHistory = []; % init the stuff
19
20        rps = randi([0, 2]); % play random first time
21        ourHistory = [ourHistory; rps];
22        return;
23    end
24
25    % update the history
26    opponentHistory = [opponentHistory; previous];
27    N = length(opponentHistory);
28
29    %extracting the features
30    f1Hist = opponentHistory(1:end-1); % f1: Opponent's move
31    ↪ 1 round ago
32    f2Hist = ourHistory(1:end-1); % f2: Our move 1 round ago
33    cHist = opponentHistory(2:end); % C: Opponent's move
34    ↪ this round
35
36    currentf1 = opponentHistory(end);
37    currentf2 = ourHistory(end);
38
39    posteriorScores = zeros(3, 1); % values fo the
40    ↪ posteriors, to be used later
41    totalSamples = length(cHist); % no of samples
42
43    for idx = 1:3 % for all: need to find Cond. probs.
44        classVal = idx - 1;
45
46        countCk = sum(cHist == classVal); % Count occurrences of
47        ↪ the target class
48
49        prior = (countCk + 1) / (totalSamples + 3); %have to add
50        ↪ +1 and +3 otherwise NaN error on clickUP grader
51        % this is caused by values being 0, so for countCk = 1,
52        ↪ totalSamples = 0 ==> prbolem!, but now will evaluate to
53        ↪ 0.667
54
55        %find cond. prob. P(f1|Ck) i.e. opponent last move
```

```

47     countf1Givenk = sum(f1Hist(cHist == classVal) == currentf1);
48     pflGivenk = (countf1Givenk + 1) / (countCk + 3);
49
50     % P(f2 | Ck) i.e. our last move
51     countf2Givenk = sum(f2Hist(cHist == classVal) == currentf2);
52     pf2Givenk = (countf2Givenk + 1) / (countCk + 3);
53
54     posteriorScores(idx) = prior * pflGivenk * pf2Givenk;      %
55     ↪ multiply for p(f|Ck)p(Ck)
56 end
57
58 [~, max_idx] = max(posteriorScores);      % for argmax part of Prob
59 predicted_opponent_move = max_idx - 1;
60
61 rps = mod(predicted_opponent_move + 1, 3);      % winning
62 ↪ move
63
64 ourHistory = [ourHistory; rps];      % update for nxt
65 ↪ iteration
66
67 % Results (10 rounds)
68 % Run      Time (s)      Rounds Won      Matches
69 ↪ Won      Anomilies
70 % 1      19.062877      648517      700
71 ↪ Only 49871 rounds for no_repeat, but still 100 matches won
72 % 2      16.313442      648523      700
73 ↪ Only 49854 rounds for no_repeat, but still 100 matches won
74 % 3      20.008488      648603      700
75 ↪ Only 49917 rounds for no_repeat, but still 100 matches won
76 % 4      22.458893      648504      700
77 ↪ Only 49870 rounds for no_repeat, but still 100 matches won
78 % 5      20.65606      648524      700
79 ↪ Only 49891 rounds for no_repeat, but still 100 matches won
80 % 6      20.963052      648418      700
81 ↪ Only 49750 rounds for no_repeat, but still 100 matches won
82 % 7      20.414442      648195      700
83 ↪ Only 49499 rounds for no_repeat, but still 100 matches won
84 % 8      18.963757      648618      700
85 ↪ Only 49985 rounds for no_repeat, but still 100 matches won
86 % 9      18.798465      648687      700
87 ↪ Only 50025 rounds for no_repeat, but still 100 matches won
88 % 10     19.613149      648338      700
89 ↪ Only 49681 rounds for no_repeat, but still 100 matches won
90 % Average 19.7252625      648492.7      700
91
92 % Results show that Naive Bayes performs relatively well for most
93 ↪ agents
94 % For non deterministic agents (no_repeat) it doesn't perform the
95 ↪ best but nevertheless all matches are still won
96 % For other agents, it can correctly predict the move that wioll
97 ↪ be played and counter it, winning 92.641814286% of rounds
98 % Compared to prac 2, Naive Bayes has better results, but took
99 ↪ longer to run and was easier to implement
100 % Compared to prac 3, NB has roughly the same results as the
101 ↪ final optimised NN, but took twice as long to run. But it was
102 ↪ much easier to code
103 % Benefit of Naive Bayes is evident: easy to implement with
104 ↪ results that are relatively good

```

84
85 end
86